Syllabus Edition
First teaching 2025
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Production Processes: Productivity & Efficiency

Exam code: 0450, 0986 & 0264, 0774

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Written by: Lisa Eades
Reviewed by: Steve Vorster
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Labour productivity

- **Production is the act of adding value to the factors of production to create goods and services** e.g. using tomatoes and basil to create a soup
  - It is the process of converting the **factors of production** into goods/services
  - It is a **measure of output** e.g. 3 cans of soup
- **Productivity is a measure of efficiency that calculates the amount of outputs produced per unit of input**
  - It calculates **how efficiently** resources are being used in the creation of goods/services and aids comparison of performance e.g. after training, workers proved to be 27% more efficient in their productivity
- **Labour productivity** is calculated using the formula

$$\text{Labour productivity} = \frac{\text{Output}}{\text{Number of workers}}$$

Worked Example

The table shows the number of pairs of luxury wool socks produced by *Sokkemani Ltd* in 2021 and 2022.

Year	Units Produced
2021	46,000
2022	69,000

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it
Help me understand

Calculate the percentage change in productivity between 2021 and 2022.

[4]

Step 1: Calculate productivity for 2021

$$= \frac{46{,}000 \text{ units}}{50 \text{ workers}} \quad (1)$$

$$= 920 \text{ units per worker}$$

Step 2: Calculate productivity for 2022

$$= \frac{69{,}000 \text{ units}}{60 \text{ workers}} \quad (1)$$

$$= 1{,}150 \text{ units per worker}$$

Step 3: Calculate the percentage difference between the two years ((new-old) / old)

$$= \frac{1{,}150 - 920 \text{ units}}{920 \text{ units}} \times 100 \quad (1)$$

$$= 25\ \%$$

Step 4: Identify whether the percentage difference is an increase or decrease

- Labour productivity has increased from 920 to 1,150 units so it is a 25% percentage **increase (1)**

Examiner Tips and Tricks

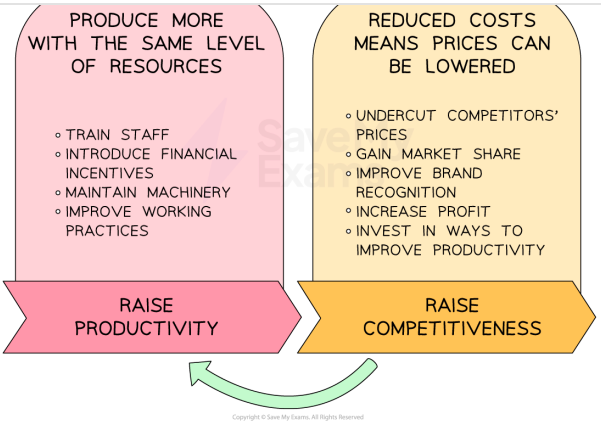
In calculation questions, always show your working. Even if the final answer is wrong, method marks can still be awarded

The importance of efficiency

- **Efficiency refers to the ability of a business to use its production resources as cost-effectively as possible**
  - Efficiency is often measured in terms of the **average cost per unit**
  - The average cost per unit is calculated using the formula

$$\text{Average cost per unit} = \frac{\text{Total costs}}{\text{Number of units}}$$

- **Maximum efficiency** is achieved when the cost per unit is at its lowest
- Businesses that increase their level of productivity (e.g of workers or capital equipment) are likely to be **more competitive**



The link between productivity and competitiveness

- Businesses that are competitive usually generate more profit
- This provides the **financial resources** to continue investing in improvements to their productivity

How to increase efficiency

- Increasing efficiency helps a business reduce costs, save time and improve productivity

Common ways to improve efficiency

1. Increasing automation

- Automation uses machines or robots to carry out tasks that would normally be done by humans.
  - Production is likely to be faster and safer, with fewer errors and lower long-term labour costs
  - For example, car manufacturers use robotic arms for welding and assembling.

2. Introducing technology

- Using new technology, such as computers or specialist software, can improve both manufacturing and service operations.
  - Technology often speeds up processes, reduces paperwork and improves communication and accuracy
  - For example, retailers using barcode scanners and stock management software to track inventory

3. Improving labour skills

- Training employees helps them work more efficiently and safely
  - They produce better quality work, with fewer mistakes and more quickly
  - For example, a café training staff to use a coffee machine correctly to avoid waste.

quality high.

- Techniques include
  - **Just-in-time (JIT)**, which involves holding less inventory to reduce storage costs
  - **Kaizen** is continuous small improvements made by employees that have an ongoing impact on operational performance

5. Better organisation of resources

- Changing the layout of a factory or office can improve workflow
  - This can save time moving between tasks or departments
  - For example, placing tools close to workers to avoid delays

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